

Assignment 1: AI System Specification

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Subject: Artificial Intelligence

Topic: Voice-Based Virtual Assistant for Multilingual Customer Support

Task 1: PEAS Specification

Performance Measure:

- **Resolution Rate:** The percentage of customer inquiries that are successfully resolved by the AI without requiring escalation to a human agent.
- **Average Handling Time (AHT):** The average time elapsed from the initial customer prompt until the final resolution or query completion.
- **Sentiment Score:** A metric derived from vocal tone analysis that measures the customer's emotional state throughout and after the interaction.
- **Language Accuracy:** The precision of the speech-to-text transcription engine across various supported languages, dialects, and accents.

Environment:

The agent operates within a web-based or phone-based customer support portal. Key external factors influencing the agent include high levels of background ambient noise, diverse regional dialects and heavy accents, latency in API connections to the backend help-desk database, and the varying complexity and ambiguity of human intent.

Actuators:

The agent utilizes a Text-to-Speech (TTS) synthesizer to provide audible, natural-sounding responses, visual display widgets (like confirmation buttons or order status cards) to guide user interaction, and secure database API interfaces to update help-desk tickets or retrieve customer account information.

Sensors:

The agent perceives its state through a high-sensitivity far-field microphone array, real-time noise-cancellation processing units that isolate voice from background noise, and digital input logs extracted from the chat or support portal.

Task 2: Environment Classification

- **Fully vs. Partially Observable:** *Partially Observable*. The agent cannot observe the user's physical surroundings or interpret nonverbal cues like micro-expressions, which frequently contain critical information about the user's emotional state.
- **Single-agent vs. Multi-agent:** *Multi-agent*. The assistant operates alongside human customers who have their own independent goals, and it often coordinates with other behind-the-scenes automated support bots.
- **Deterministic vs. Stochastic:** *Stochastic*. Due to the inherent variability of human speech patterns, complex emotions, and network traffic, the outcome of any given interaction cannot be predicted with absolute certainty.
- **Episodic vs. Sequential:** *Sequential*. Every response provided by the agent is heavily dependent on the context of the conversation and the history of previous interactions, making it a stateful process.
- **Static vs. Dynamic:** *Dynamic*. The conversation environment is constantly shifting; user sentiment can evolve rapidly in real-time, and external noise conditions change as the user speaks.
- **Discrete vs. Continuous:** *Continuous*. Speech recognition requires processing continuous audio waveforms and natural language probabilities rather than a finite set of discrete states.

Task 3: Critical Analysis

1. **Greatest Technical Challenge:** The *Partially Observable* dimension is the most significant hurdle. Since the assistant is voice-only, it must accurately infer user intent and emotional nuance purely through acoustic parameters (pitch, tempo, tone). Misinterpreting an ironic or frustrated tone can lead to catastrophic failures in customer service, requiring highly complex emotional recognition models to solve.

2. **Utility Function:** A balanced utility function could be: $U = 0.6 * (Resolution_Quality) - 0.4 * (Wait_Time)$. This formula balances the need for perfect accuracy against the user's finite patience. If the weight of "Resolution_Quality" were doubled, the system would shift its behavior to prioritize precision and clarity, potentially conducting extra validation steps to ensure the answer is correct, even if it adds to the overall wait time.

Structured Representation

```
{  "system_name": "Voice-Based Virtual Assistant for
Multilingual
Support",
  "peas": {
    "performance_measures": [
      "resolution_rate",
      "average_handling_time",
      "sentiment_score",
      "language_accuracy"
    ],
    "environment": "Customer support portal with
variable audio noise and multi-language input",
    "actuators": ["text_to_speech", "ui_widgets",
"ticket_database_api"],
    "sensors": ["far_field_microphones",
"noise_cancellation_processor", "log_inputs"]
  },
  "environment_classification": {
    "observability": {
      "choice": "Partially Observable",
      "justification": "The agent lacks access to the user's
```

```

visual context and non-verbal body language."
},
  "agents": {
    "choice": "Multi-agent",      "justification":
"The system acts alongside human customers and other
collaborative software bots."    },
    "determinism": {
      "choice": "Stochastic",      "justification": "Human
intent and environmental noise are unpredictable variables."
    },
    "episodic": {
      "choice": "Sequential",      "justification": "The
interaction is stateful, where every word depends on the
previous turns."                  },
    "static": {
      "choice": "Dynamic",         "justification": "The
conversation environment changes constantly with voice
input and sentiment shifts."      },
    "discrete": {
      "choice": "Continuous",      "justification": "Audio
signals and language intent are modeled on continuous
probabilistic values."            }
  },
  "utility_function": "U = 0.6 * resolution_quality -
0.4 * wait_time"
}

```